HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Owned and managed assay requirements throughout the product lifecycle, translating customer, clinical, and business needs into traceable technical requirements for **diagnostic solutions**.
- Served as the primary technical liaison between **Assay Development** and **Analyser Engineering** teams, facilitating alignment across scientific, engineering, and systems development stakeholders.
- Defined integration strategies for assays with **software**, **consumables**, **sample preparation**, and **LC-MS instrumentation** to deliver complete diagnostic systems.
- Collaborated with **Quality**, **Regulatory**, and **Risk Management** teams to ensure adherence to applicable **IVD standards** and compliance objectives.
- Designed and executed verification and validation activities, ensuring integrated assay solutions met performance specifications and readiness for **commercial deployment**.
- Developed Python scripts to automate data ingestion and transformation from laboratory instruments, reducing manual processing time by **~40%** and improving data accuracy for downstream analysis.
- Containerized analytical applications using **Docker** and managed deployments on **AWS EC2** instances, establishing CI/CD pipelines with **Jenkins** for streamlined code integration and release cycles.

SKILLS

Domain – Assay Development, Product Development Processes, IVD Standards, Diagnostic Workflows, Life Sciences, Medical Device Industry

Systems Engineering – Systems Requirements, Technical Requirements, System Trade-offs, Performance Requirements, Verification, Validation

Collaboration & Communication – Multidisciplinary Teams, Stakeholder Management, Technical Decision-Making, Problem-Solving, Communication Skills

Programming & Scripting – Python, Java, SQL, Bash Scripting

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Analysis & Tools – Power BI, Tableau, R, pandas, NumPy, scikit-learn, DAX, Power Query

PROJECTS

Assay Integration Platform Development

- Designed a modular Python-based framework to integrate data from various analytical instruments, standardizing input formats for a central data repository.
- Implemented automated data validation checks using pandas and SQL queries, identifying and flagging over **15%** of data anomalies during ingestion.
- Developed shell scripts for deployment and configuration management of the integration services across development and staging environments.

Automated Reporting for Clinical Data

- Built interactive **Power BI dashboards** to visualize key performance indicators and assay results for clinical stakeholders, consolidating data from multiple sources.
- Developed **DAX measures** and Power Query transformations to enable drill-down capabilities and comparative analysis of historical and current assay runs.
- Collaborated with laboratory scientists to gather requirements and refine dashboard layouts, ensuring actionable insights were clearly presented for decision-making.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)